

3D ICE: An Interactive Browser-Based Cryosphere Explorer for Antarctica and Greenland

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Software

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Summary

3D ICE is an open-source platform for exploring research-grade representations of the Antarctic and Greenland ice sheets in a web browser (Figure 1). It combines terrain, ice velocity, ocean circulation, ice-shelf basal melting, basal friction, subglacial hydrology, and drainage basins in layered three-dimensional scenes. Users can rotate and zoom each ice sheet, switch datasets and quality levels, search polar places and geographic features, show the ice-free bed after isostatic rebound, and follow links to the original data products.

An offline Python pipeline converts heterogeneous NetCDF, HDF5, GeoTIFF, and vector products into compact binary arrays paired with machine-readable provenance metadata. A static JavaScript application, built with Three.js (Cabello & three.js contributors, 2024) and WebGL, decodes these packages and renders them without plugins or server-side computation. This separation keeps the scientific transformations reproducible and the public interface simple to deploy, share, and use on desktop and mobile devices.

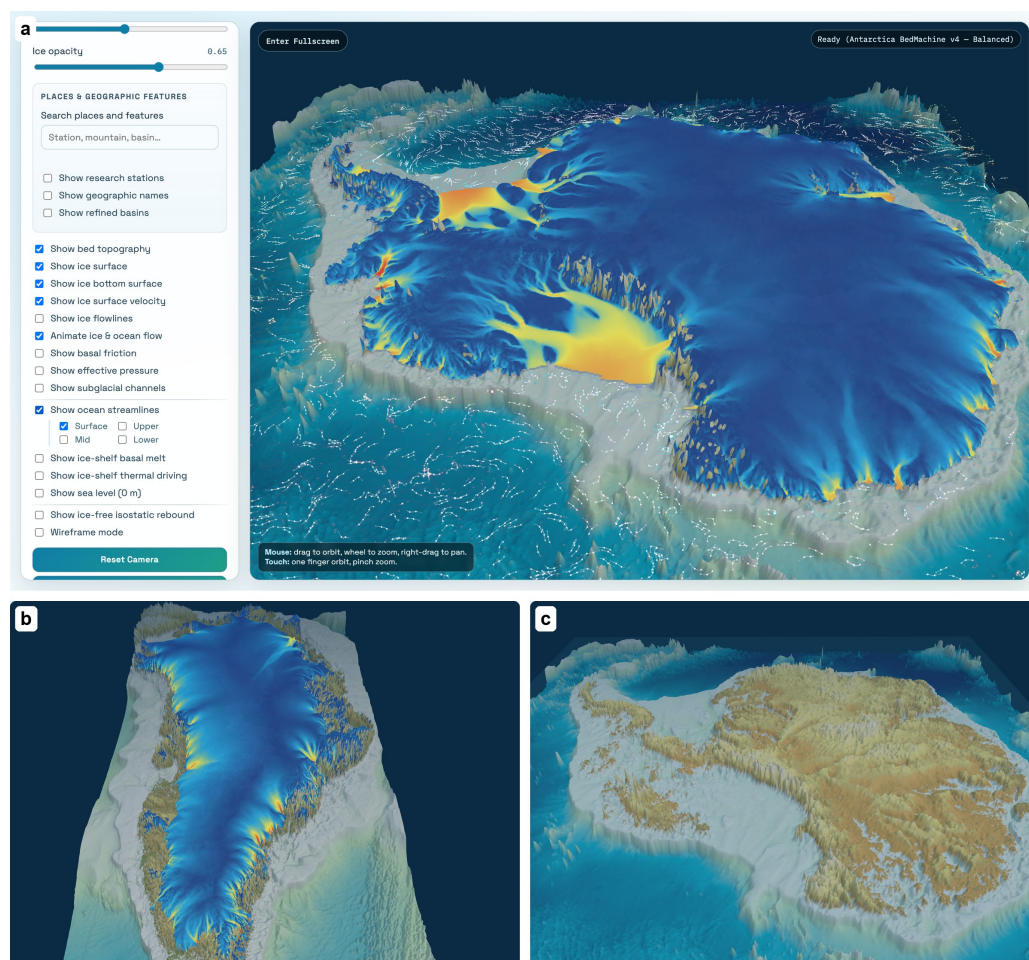


Figure 1: The 3D ICE explorer. (a) The interface, with Antarctic ice-surface speed (Mouginot et al., 2019) and surface-layer WAOM2 ocean streamlines (Richter et al., 2022) over BedMachine Antarctica v4 (Morlighem et al., 2020). (b) Greenland ice-surface speed from ITS_LIVE (Gardner et al., 2024) over BedMachine Greenland v6 (Morlighem et al., 2017). (c) The ice-free Antarctic bed after complete isostatic rebound under the idealised regional-flexure response. Vertical scales are exaggerated.

Statement of need

Modern cryosphere science generates large, gridded datasets in specialized formats and polar stereographic coordinate systems. Inspecting several products together commonly requires a desktop GIS, a numerical computing environment, or purpose-built scripts. These workflows suit quantitative analysis but require installation, data transfer, reprojection, and preprocessing before a user can see an ice sheet in context.

That barrier particularly affects researchers in adjacent disciplines, educators who want to use real polar data in teaching, and scientists preparing interactive research communication. They need a rapid way to see relationships among ice geometry, flow, ocean forcing, and basal processes while retaining a path back to the source datasets. 3D ICE addresses this need with a curated, zero-install view rather than attempting to replace GIS or numerical analysis. Its research purpose is to make heterogeneous cryosphere products easier to compare, explain, and select for subsequent analysis.

State of the field

Quantarctica is a comprehensive Antarctic data package, analysis environment, and visualization platform centred on a desktop QGIS workflow (Matsuoka et al., 2021), and QGreenland is its counterpart QGIS data package for Greenland (Moon et al., 2023). Both are well suited to local geospatial analysis and span a broader range of disciplines than 3D ICE. In contrast, 3D ICE trades general GIS operations for an immediately shareable, curated 3D experience that uses a common interaction model for both Antarctica and Greenland.

NASA Worldview provides rapid web access to more than a thousand global satellite-imagery layers, including polar views, temporal comparison, animation, and data download (NASA Earth Science Data and Information System, 2026). Its focus is two-dimensional, often near-real-time Earth observation imagery. 3D ICE instead combines terrain with subsurface and model-derived fields such as basal friction, hydrology, basal melt, and depth-dependent ocean circulation.

General 3D geospatial libraries such as CesiumJS (Cesium GS, Inc. and CesiumJS contributors, 2026) provide a high-precision WGS84 globe, scalable data formats, and rendering primitives for building web applications. Three.js provides the lower-level graphics foundation used by 3D ICE (Cabello & three.js contributors, 2024). Neither library supplies a cryosphere data model, polar-grid conversion, provenance records, layer semantics, or data curation. Contributing these product-specific transformations to a general rendering engine would therefore not address the research need. 3D ICE reuses established graphics infrastructure while concentrating its scholarly contribution in the reproducible polar-data pipeline, explicit data contract, and domain-specific interaction design.

Software design

3D ICE uses a two-stage, contract-oriented architecture. The offline stage reads authoritative source products, reconciles their coordinate conventions, and resamples fields onto terrain-aligned polar grids with NumPy (Harris et al., 2020). Floating-point grids are quantized to signed 16-bit arrays for browser delivery. This introduces a controlled precision-versus-transfer-size trade-off: scale, offset, fill value, units, grid geometry, statistics, source citation, and processing provenance are stored in a paired .meta.json file so that decoding is deterministic and the loss of precision is inspectable. The metadata schema, specified in the repository, is the boundary between scientific preparation and visualization. A single decoder module reads it in the page, in the geometry worker, and in a test that decodes every committed package and checks it against the statistics the pipeline recorded before quantizing: every quantized field reproduces them to within half a quantization step.

Computationally expensive transformations are performed before deployment. For example, Antarctic ocean-current packages are generated by advecting particles through multi-depth WAOM2 velocity fields, balancing seeds by region and depth, and controlling streamline density (Boeira Dias et al., 2023; Richter et al., 2022). This offline choice leaves the public application needing only static hosting, with no database, application server, or restricted compute resources. The trade-off is that integrating or updating a source product requires regenerating and versioning its web package.

The online stage fetches binary arrays, reconstructs terrain and overlays, and renders them with Three.js. Geometry construction runs in Web Workers so that large grids do not block interface updates. Balanced and HD packages make the resolution-versus-memory choice explicit across mobile and desktop devices. The English and Chinese pages are thin shells around one shared runtime; logic that needs neither the page nor the scene, such as package decoding, place search, label styling, and refined-basin validation, lives in separate modules that run unchanged under Node's test runner. URL parameters select the region, terrain dataset, and display mode, and every data layer links to its source product.

The isostatic-rebound layer shows which parts of today's sub-sea-level bed would emerge once glacial isostatic adjustment is complete. By default it displays the published response of (G. Paxman et al., 2026; G. J. G. Paxman et al.; 2022), sampled at the terrain nodes: elastic-plate flexure with laterally variable elastic thickness (Steffen et al., 2018; Swain & Kirby, 2021), remaining post-LGM disequilibrium, and water loading under a sea surface raised by both ice sheets' meltwater. For comparison it also solves an elastic-lithosphere, relaxed-asthenosphere steady state (Le Meur & Huybrechts, 1996; Lingle & Clark, 1985) spectrally (Bueler et al., 2007) and, as an upper bound on peak uplift, local Airy isostasy (Turcotte & Schubert, 2002), with an adjustable datum; the solver is verified against the analytic point-load solution in Kelvin functions (Brochie & Silvester, 1969). Sea-level-equivalent figures follow Gregory et al. (2019). The interface states each response's assumptions, including that the present bed is not in balance with the present load (Whitehouse et al., 2019) and that low mantle viscosity beneath parts of West Antarctica shortens the response time by orders of magnitude (Barletta et al., 2018).

The domain contract is checked at several levels: unit tests cover quantization, coordinate sampling, metadata statistics, search, the rebound solver, and bounded geometry; the contract test covers every committed package; integration tests regenerate derived packages and compare them with the committed files; compatibility tests build a distributable static bundle; and Playwright tests load the explorer and exercise browser interactions. These checks run in continuous integration and locally, without the production website.

Research impact statement

Y. Wang uses 3D ICE to compare candidate bed, velocity, and basal-friction products before configuring ice-flow and subglacial-hydrology simulations, and has used its views in a research funding proposal. It has been demonstrated in talks at the Antarctic Research Centre, Victoria University of Wellington (February 2026), the ISMIP7 Workshop in Copenhagen (March 2026, presented by C. Zhao), the Asia Early Career Polar Forum in Zhuhai (June 2026), and the School of Oceanography, Shanghai Jiao Tong University (July 2026).

Y. Lin has used 3D ICE in teaching GE1301 Climate Change and Extreme Weather at the City University of Hong Kong, to show students the geography, ice flow, and surrounding ocean of Antarctica and Greenland, and is scheduled to demonstrate it to the public at InnoCarnival 2026 (Hong Kong Science Park, October–November 2026). F. Boeira Dias, who provided the WAOM2 output that 3D ICE renders, has described using the explorer to visualise ocean circulation that is difficult to interpret in standard two-dimensional plots (Australian Antarctic Program Partnership, 2026; Spatial Source, 2026). From its launch on 22 March to 27 September 2026, the public site recorded 578 active users and about 1,200 page views.

The public application and versioned compatibility bundle (Wang & Lin, 2026) integrate representative products for bed geometry (Morlighem et al., 2017, 2020; Palmer et al., 2025; Pritchard et al., 2025), ice velocity (Gardner et al., 2018, 2024; Mouginot et al., 2019), ocean circulation (Boeira Dias et al., 2023; Copernicus Marine Service, 2025; Richter et al., 2022), ice-shelf basal melting (Galton-Fenzi et al., 2025), basal friction (Jager, Gladstone, Zwinger, Wang, et al., 2026; Jager, Gladstone, Zwinger, Gillet-Chaulet, et al., 2026; Jager, 2026a, 2026b), subglacial hydrology (Ehrenfeucht, 2024; Ehrenfeucht et al., 2025; Werder et al., 2013), drainage boundaries (Mouginot et al., 2017; Mouginot & Rignot, 2019; Rignot et al., 2011), and polar places (Council of Managers of National Antarctic Programs, 2024; Scientific Committee on Antarctic Research, 2026).

The repository supplies a specified data contract, a documented preparation pipeline, tagged releases with a changelog, contribution and support pathways, and worked examples that run from a clean clone. One reproduces the rebound figures the explorer reports for Antarctica, including 2.94 million km² of today's sub-sea-level bed emerging above the ice-free sea surface; another regenerates six derived data packages whose payloads match the committed files byte

for byte.

AI usage disclosure

Generative AI was used extensively in developing 3D ICE. The first prototype of the explorer, and much of its subsequent application code, was generated with the OpenAI Codex coding agent (GPT-5-family models, February to August 2026) from Y. Wang's natural-language specifications, data, and review, as the project's public coverage describes ([Australian Antarctic Program Partnership, 2026](#)). Anthropic's Claude Code (Claude Opus 4.6 in March and April 2026; Claude Opus 5 and Claude Opus 5.5 in September 2026) was used for later work: parts of the interface, test infrastructure, the isostatic-rebound layer and its data pipeline, the refactoring that moved the runtime into shared and tested modules, the repository documentation, reference checking, and the drafting and copy-editing of this paper. Y. Wang selected the datasets and scientific methods and set the architecture. Every AI-generated step was checked by hand before it was kept: Y. Wang read each change, ran the explorer to confirm its behaviour, validated each scientific layer against its source product, and checked every reference against its source. Automated benchmarks back this review: every decoded package is compared with the statistics recorded before quantization, and the rebound solver with the analytic point-load solution. The authors accept responsibility for the accuracy, originality, licensing, and integrity of the submitted work.

Author contributions

Y. Wang conceived, designed, and developed 3D ICE and wrote the paper. Y. Lin helped debug and evaluate the explorer, advised on the isostatic-rebound and sea-level calculations and on the interface design for public audiences, and joined discussions of its development.

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